

Prediction of the Structural Properties of MOFs with High Hydrogen Storage Capacities at Room Temperature and Moderate Pressures using Neural Networks

Inverse Design of Metal–Organic Frameworks for Hydrogen Storage at Room Temperature and Moderate Pressure Using Neural Networks

A. Granja-DelRío^{a,*}, I. Cabria^a, T. Calonge^b

^a*Departamento de Física Teórica, Atómica y Óptica, Universidad de Valladolid, ES-47011 Valladolid, Spain*

^b*Departamento de Informática (ATC, CCIA y LSI), Universidad de Valladolid, ES-47011 Valladolid, Spain*

Abstract

Metal-Organic Frameworks (MOFs) are promising solid porous materials for hydrogen storage at room temperature and low or moderate pressures for Fuel Cell Vehicles (FCVs). Study and exploration of the hydrogen storage capacities of MOFs and other solid porous materials using Grand Canonical Monte Carlo (GCMC), Density Functional Theory (DFT) and other methods are computationally expensive. An Artificial Neural Network (ANN) approach is presented and trained on a dataset of 1000 MOFs, obtained from GCMC simulations of the MOFs at 298.15 K and 25 MPa. More precisely, each sample contains the hydrogen usable gravimetric and volumetric capacities at that temperature and pressure and five structural properties of the 1000 MOFs. The aim of the ANN is to find the Structural Properties of one or more MOFs whose hydrogen usable volumetric and gravimetric storage capacities at 298.15 K and 25 MPa are consistent with the Department of Energy (DOE) hydrogen storage targets for 2025. In practice, the problem formulation is just the opposite: Use Machine Learning (ML) methods to find/obtain the hydrogen usable storage capacities of MOFs, using a dataset of known MOFs. The ANN is used to find an input space region (structural properties region) where a solution (a MOF with given hydrogen usable gravimetric and volumetric capacities at 298.15 K and 25 MPa equal to the DOE targets) could be searched for. GCMC simulations of some MOFs close to the found input space region of structural properties have been also performed. The usable gravimetric capacities are close to the given capacities, but

*Corresponding author

Email address: alejandra.granja@uva.es (A. Granja-DelRío)

~~not to the usable volumetric capacities.~~ Unlike conventional Machine Learning (ML) algorithms that directly predict hydrogen uptake from structural descriptors, the present work formulates the problem in an inverse design manner, using an ANN to identify regions of the structural-property space compatible with target storage performances. This strategy enables the rapid screening of candidate MOFs and provides physically interpretable guidance for materials discovery.

1. Introduction

Transport is essential for modern life. It connects people and supports trade, culture, and access to basic services. However, it also has strong negative impacts. It is responsible for about 25 % of greenhouse gas emissions in the EU [1, 2, 3], and it increases air and noise pollution as well as habitat loss [4]. For this reason, cleaner energy solutions are needed.

Hydrogen is seen as a good alternative for reducing emissions. In vehicles, it can replace fossil fuels, especially in Fuel Cell Vehicles (FCVs). Its gravimetric energy density is very high, 120 MJ/kg, while gasoline offers only 44 MJ/kg [5]. However, the volumetric energy density is low at ambient conditions and moderate pressures (0.5–25 or 35 MPa). This makes onboard storage a difficult challenge. Current systems compress hydrogen up to 70 MPa [6], ~~which increases costs and safety concerns.~~ **which is common in commercial compressed-gas storage technologies, but increases cost, system complexity, and safety requirements.**

To face this problem, the US Department of Energy (DOE) defined targets for 2025. They require at least 5.5 wt. % for the usable gravimetric capacity, ug_c , and 0.040 kg/L for the usable volumetric capacity, ug_v [5, 7]. These values are considered essential for real applications, where driving ranges above 600 km are needed [8, 9, 10, 11, 12].

Among possible storage materials, Metal-Organic Frameworks (MOFs) are very promising. They combine high surface area, tunable pore sizes, and flexible design. This makes them good candidates for hydrogen storage near room temperature through physisorption [13]. Still, finding new MOFs that meet both gravimetric and volumetric targets is not straightforward. It often requires extensive experimental work or computationally demanding simulations such as Grand Canonical Monte Carlo (GCMC) and Density Functional Theory (DFT) [14, 15].

In recent years, ~~machine-learning~~ ML techniques have been extensively applied to accelerate the prediction of hydrogen storage properties in MOFs and other porous materials. Previous studies have explored the use of ~~artificial neural networks~~ Artificial Intelligence systems like random forests,

and other regression models to predict hydrogen uptake or deliverable capacities from structural descriptors [16, 17, 18, 19, 20, 21, 17]. These approaches have demonstrated that data-driven models can significantly reduce the computational cost associated with high-throughput GCMC or DFT simulations. ~~Qué previous?~~

However, most existing studies focus on the direct prediction of hydrogen storage performance from known structural features, effectively addressing a forward-design problem. In contrast, less attention has been paid to the inverse formulation, where desired storage targets are used to infer regions of the structural-property space that could host suitable materials. This limitation restricts the direct applicability of ~~machine-learning~~ ML models as design-guidance tools for experimental synthesis.

Tens of thousands of synthesized MOFs and millions of theoretical and not synthesized MOFs have been reported [22, 23]. Their large diversity and the use of different metals give researchers many options to design new structures with high potential for hydrogen storage.

~~From a mathematical point of view, a search in a solutions space must be taken place, in order to find materials to construct an adequate Hydrogen-Cell Vehicle.~~ From a computational perspective, the identification of suitable MOFs for onboard hydrogen storage can be formulated as a search problem within a high-dimensional materials space. To do that, a MOFs table will be available. Each row will store the characteristic parameters of a given material. The mentioned GCMC will provide the corresponding u_{g_c} and u_{g_c} , as it will be described in the following section. All this information will make up the dataset for any ~~Machine Learning~~ ML algorithm, in this work, used as regression system, also know as function estimator [24]. In particular, an approach based on ANN will be presented, but previously other ML function estimators have been tested and discarded such as decision tree or support vector machine based prototypes.

The ANN will be trained using a dataset composed by the usable storage capacities at 298.15 K and 25 MPa plus five structural properties of 1000 MOFs. The usable capacities were obtained by means of GCMC simulations performed using an in-house code published in GitHub [25]. When ANNs are applied in the field of materials science, the usual procedure is as follows: the structural parameters of the materials are the inputs and the desired capabilities of the materials are the outputs. ~~In this work, the problem statement will be the opposite:~~ In this work, an alternative strategy is explored, for given u_{g_c} and u_{g_c} (given/desired properties), ~~the ANN will obtain the five structural parameters of an hypothetical MOF~~ the ANN estimates the ranges of eighth structural

parameters of a hypothetical MOF whose usable storage capacities will be equal or larger than the DOE capacity targets for 2025. Finally, GCMC simulations of MOFs whose structural properties are close to the structural properties predicted by the ANN, will be also performed and analyzed.

The main novelty of the present work lies in formulating hydrogen storage optimization in MOFs as an inverse-design problem at near-ambient temperature and moderate pressure. Instead of predicting storage capacities from given structures, the ANN is used to identify regions of the structural-parameter space that are consistent with DOE performance targets. This shift from forward prediction to inverse mapping provides actionable guidance for materials design and complements existing high-throughput screening methodologies.

This work builds upon previous data-driven studies while explicitly positioning the ANN as a complementary tool to conventional high-throughput screening approaches rather than as a replacement for detailed molecular simulations.

2. Methodology

2.1. Grand Canonical Monte Carlo Simulations

Grand Canonical Monte Carlo (GCMC) simulations were performed to evaluate the total and usable hydrogen gravimetric and volumetric storage capacities of MOFs chosen from the Cambridge Crystallographic Database Centre (CCDC) [26]. The simulations were carried out at 298.15 K with pressures at 0.5 and 25 MPa.

Each GCMC run consisted of ten millions steps. The first five millions were used to reach equilibrium, and the last five millions were employed to calculate the storage values. The Metropolis algorithm was applied during all iterations [27]. The types of moves on each iteration were assigned with the following probabilities, after running some tests: 20 % for molecular displacement, 40 % for deletion, and 40 % for insertion [28, 13, 29].

The chemical potential was determined with the Soave–Redlich–Kwong (SRK) equation of state [30]. The values for the acentric factor ω , the critical pressure P_c , and the critical temperature T_c of hydrogen and methane were taken from Zhou and Zhou [31] and Xu et al. [32].

The interactions between the framework atoms and the gas molecules were described with Lennard–Jones (LJ) potentials [33]. The LJ parameters were calculated using the Good-Hope combining rule for σ [34] and the Berthelot rule for ϵ [35]. Specific LJ parameters were used for

the C and H₂ interaction, without applying the Berthelot-Good Hope combining rule. The values of the LJ parameters of the atoms and molecules used in the simulations can be found in GitHub [25]. The cutoff radius was set to 20 Å, after running several tests [28, 13, 29]. In the GCMC simulations, the atoms that composed the MOFs were treated as rigid.

All simulations were executed using an in-house program, published in GitHub and written in fortran [25]. The in-house GCMC code employed in this work has been previously validated against both experimental measurements and results obtained with established simulation packages, including RASPA, for hydrogen adsorption in benchmark MOFs. In particular, systematic comparisons for materials such as IRMOF-1, IRMOF-20, and ZIF-8 under a wide range of temperatures and pressures have been reported in previous studies, showing good agreement in both gravimetric and volumetric storage capacities [29, 36, 37, 38].

These validation studies demonstrate that the present implementation reliably reproduces the magnitude and trends of hydrogen adsorption reported in the literature. The use of in-house GCMC codes for large-scale screening has also been adopted in earlier works, as it provides greater flexibility and reduced computational cost when exploring high-dimensional materials spaces. While widely used software packages such as RASPA [39] remain a reference tool for adsorption simulations, the present implementation offers comparable accuracy for the purposes of comparative screening and inverse property mapping considered in this work.

The input of the program for each MOF consisted mainly on the cell of the MOF, obtained from the CIF (Crystallographic Information File) file on the CCDC database. Part of the input was also the number of iterations, the probabilities of the types of moves, the type of equation state (SRK in the present case) and the cutoff radius. The program calculates the total and usable volumetric and gravimetric storage capacities, which are defined in the next paragraphs.

Prior to the GCMC simulations, all MOF structures obtained from the CCDC were carefully prepared for computational use. Solvent molecules and non-framework species were removed from the CIF files, and the structural integrity of the frameworks was verified by checking atomic positions and unit cell parameters. As a result, all structures used in this work are computation-ready and equivalent in quality to those provided by curated databases such as the CoRE MOF database.

The five structural parameters of each MOF were obtained by means of the Zeo++ code [40]. Those parameters were: the density in kg/L, the porosity (a dimensionless magnitude; its values are between 0 and 1), the radius of the largest pore of the MOF, R_i , in Å, the specific surface area,

SSA, in m^2/g , and the specific pore volume, SPV, in cm^3/g .

2.1.1. Definitions of Storage Capacities

In these simulations, the total volumetric storage capacity of hydrogen gas at pressure P and temperature T , $v_c(P, T)$, also referred to as the density of stored gas, is expressed as

$$v_c(P, T) = \frac{M_g(P, T)}{V} , \quad (1)$$

where $M_g(P, T)$ corresponds to the total mass of gas (hydrogen) stored inside the simulation box at pressure P and temperature T , and V denotes the box volume. In this work, v_c is reported in units of kg of H_2/L .

The total gravimetric storage capacity at pressure P and temperature T , $g_c(P, T)$, is determined from

$$g_c(P, T) = \frac{100M_g(P, T)}{M_g(P, T) + M_{ads}} , \quad (2)$$

with M_{ads} represents the mass of the porous adsorbent material contained in the simulation cell, respectively. The total g_c is expressed in weight percent (wt. %).

The usable gas mass at P and T , $M_{ug}(P, T)$, is defined as the difference between the total gas mass stored at a given pressure P and temperature T and the gas mass that remains at the lower (depletion) pressure, P_{dep} , at the same T , $M_{ug}(P, T) = M_g(P, T) - M_g(P_{dep}, T)$. **In this work, the depletion pressure was set to $P_{dep} = 0.5$ MPa, consistent with values used in the literature for defining usable or working hydrogen storage capacities in adsorption systems. This choice corresponds to the lower bound of pressures accessible in typical fuel cell operation [15, 41].** The usable volumetric and gravimetric capacities, also called working or deliverable capacities [5, 7, 42], follow directly from Eqs. 1 and 2, respectively, replacing the total mass, $M_g(P, T)$, by the usable mass, $M_{ug}(P, T)$, into the definitions. The usable and total capacities have the same corresponding units: gravimetric in wt. % and volumetric in kg/L . In this study, only the usable storage capacities are considered and discussed.

2.2. Dataset

~~The ANN~~ Several ML systems were considered and implemented using a dataset of 1000 MOFs, which was structured as a table with seven columns and 1000 rows, was generated from GCMC sim-

ulations at 0.5 MPa and 25 MPa performed on MOFs randomly selected from the CCDC database. Each row represents one MOF and contains: the two usable capacities (u_g in wt.% and u_g in kg/L at 298.15 K and 25 MPa) and the five basic structural parameters. In this work, the ANN takes the two usable capacities as inputs and predicts the five structural magnitudes as outputs.

2.3. Machine Learning Regression

Considero que en metodología solo se definen los métodos, no se hacen pruebas para seleccionarlo, esta parte va en resultados, para decir por qué se coge uno u otro

ML Theory originally formulated two fundamental supervised learning challenges: classification and regression [43, 44]. From an external data point of view, the difference between them lies simply in the numerical type of the desired output associated with each input. In effect, for classification problems, the output needs to be discrete while for regression, it must be continuous [45]. From decision trees up to some of approaches belonging to the large group called "Linear Methods" have both classification and regression version. Without a doubt, the most relevant examples for that application are in the ANN category, since they could simulated a wide variety of functions even if a high no linearity will be present [46].

Before model training, an experimental protocol and evaluation metric were defined. A Hold-Out strategy with a 2/3–1/3 split between training and testing sets was adopted as an initial validation procedure [43]. The coefficient of determination, R^2 , was selected as the primary regression metric, as it provides a direct measure of the proportion of variance explained by the model [47]. However, another metrics were used like Mean Absolute Error (MAE), Root Mean Square Error (RMSE), as well as the Mean Squared Error (MSE) also used as criterion for optimization process in several training models as the mentioned ANN in the previous paragraph.

In order to define an appropriate regression framework for the present problem, several widely used supervised learning regressors were considered, following standard practice in the literature [46]:

- Decision Tree Regressor (DTR)
- Support Vector Machine with a linear kernel (SVR)
- Multilayer Perceptron (MLP-R)

The first one consists of a decision tree based on an optimized version of the CART (Classification and Regression Trees) algorithm as implemented by SCIKIT-LEARN[®]. The second one is a support vector machine employing a linear kernel. The SCIKIT-LEARN[®] documentation specifically recommends this kernel type for large-scale datasets, which corresponds to the present case. Moreover, several studies corroborate this methodological choice [48][49]. Both prototypes were programmed in Python[®] using the appropriate SCIKIT-LEARN[®] module. Due to their implementation requirements, it was necessary to handle a list of individual regressor: one for each output. Each list of these ML regressor was integrated into a single object, for which the typical methods (`__init__`, `fit`, `predict`, and `score`) were implemented.

Regarding the last model (MLP-R), in contrast to the previous ones, its architecture allows considering all outputs as inter-correlated, which will be discussed later. At the programming level, it was opted for direct Pytorch[®] implementation rather than using the corresponding module SCIKIT-LEARN[®]. This decision was driven by Pytorch[®]'s superior flexibility, which allowed us to experiment with different activation functions across distinct layers [50].

2.4. Multilayer Perceptron Regressor

As will be demonstrated in the subsequent sections, the multilayer perceptron is proposed as the final prototype. The Fig. 1 shows the architecture of the mlp used in this work, that consists of three hidden layers of equal size and an output layer. A priori, one might consider applying the identity function, given that this constitutes a regression problem. However, since the output must be interpreted as a physical variable constrained within specific margins, a bounded activation function is required. Furthermore, as the output cannot be negative in any case, the sigmoid function emerges as a suitable candidate, not least because it offers the advantage of being differentiable [51]. The use of this function will subsequently necessitate a corresponding scaling operation, based on the predefined maximum and minimum limits for the magnitude represented by each output. This scaling, and its inverse, must be applied beforehand, given that neural networks typically require their input numerical values to be normalized within the $[0, 1]$ interval.

Regarding the activation function for the hidden layers, the rectified linear unit (ReLU) constitutes a suitable choice. This is motivated by two primary considerations. First, ReLU preserves the bounded nature of the input values. Second, it helps to minimize the vanishing gradient phenomenon, which is particularly relevant in deep architectures such as the four-layer network

presented herein [52].

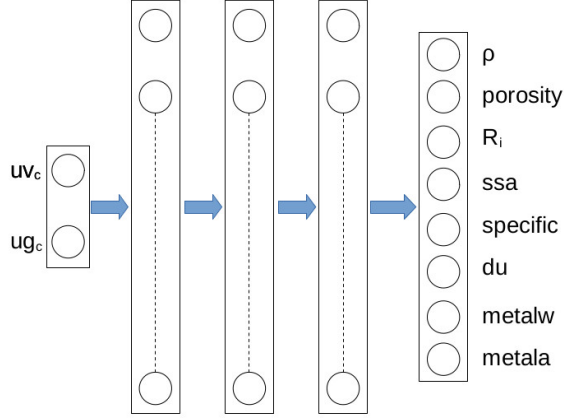


Fig. 1. MLP Regressor Architecture

After the hyperparameter exploration stage, the selected MLP architecture is retrained using the full dataset. In this final training step, all available samples are combined to build a single regression model, ensuring that the maximum amount of information is used. This procedure is standard practice once model configuration has been fixed, and it avoids unnecessary loss of training data due to validation splits.

The rest of implementation details correspond to computation topics. First of all is the programming language and the interactive development environment (IDE). In this particular case, both Python[®] and Jupyter[®] were chosen. However, the most relevant tool for the ANN was Pytorch[®], which has been built upon tensor calculus to enable an efficient and flexible Deep Learning computation. All the proposed model programs were prepared to be executed automatically on both CPU and GPU; the last option only works if a corresponding hardware device is available and compatible with the CUDA^{®1} or ROCm^{®2} version used by Pytorch[®].

Under this configuration, the system must tune a total of 968 trainable parameters, considering that each hidden layer consists of 10 neurons. On the other hand, given that there are 4,500 training samples corresponding to 9 out of 10 cross-validation folds (see next paragraph), and each sample has two attributes (inputs), the risk of underfitting is avoided. This is because the inputs-to-weights

¹Compute Unified Device Architecture - NVIDIA's parallel computing platform and programming model.

²ROCm[®]: AMD[®]'s open-source alternative to NVIDIA's CUDA[®].

ratio is close to 10, which is more than sufficient [53].

Regarding the experimental methodology, Cross-Validation was chosen instead of the Hold-Out method used in the preceding subsection. Hold-Out was adequate there because only a single numerical value was required for preliminary comparison and model screening. However, to draw reliable conclusions, a single Hold-Out split is insufficient; multiple experiments with varying train/test splits are necessary. Repeated Hold-Out could be a viable solution, as the variance of the estimate does not depend on a single split but is distributed across multiple experiments. Cross-Validation offers a similar advantage but with a key benefit: it ensures that all samples are used for both training and testing. This property cannot be guaranteed by Repeated Hold-Out [47].

Previously, each input and output was scaled into the interval $[0,1]$. In this step, there is only one thing left to do with samples: organizing them into batches. With the aim to avoid any rote learning, the samples will be permuted in each epoch. As it is well known, this task is the foundation for every Stochastic Optimization process, that will also improve the power of generalization, which is the main goal of any machine learning system [54].

To execute the algorithm, it is necessary to set some optimization issues. The most important is the cost function. In a regression problem, the Mean Squared Error (MSE) is a very good candidate for assessing optimization and performance [44]. The next step consists on the optimizer election according to the preceding lost function. The Adaptive Moment Estimation (ADAM) is probably the most popular ANN optimizer [55]. It is based on gradients that include not just the momentum term in an adaptive manner, but also incorporates several regularization algorithms like L1 and L2 [56].

The model implementation was split into two phases:

- Hyperparameters setting
- Production mode

The first methodology requires extensive trial-and-error experimentation, guided by empirical rules, to identify suitable MLP hidden layer architectures and optimizer hyperparameters (e.g., learning rate, momentum, and regularization coefficients). This represents the most computationally expensive phase, as the number of trials is multiplied by the number of cross-validation folds (10).

The second phase, whose name is borrowed from Software Engineering, corresponds to the

deployment stage, in which the system is delivered to the client and is assumed to be ready for operation by the target user. At this point, parameter tuning is carried out by exploiting the most comprehensive understanding of the problem domain. This implies there will be no samples reserved for evaluation, as this process has already been executed in the previous phase. Therefore, all available samples are used for training.

3. Results and Discussion

Table 1 shows different metrics for evaluating the systems tested in this work:

- MAE: Mean Absolute Error
- R^2 : R-squared
- MSE: Mean Squared Error
- RMSE: Root Mean Squared Error

ML regressor:	DTR	SVR	MLP-R
MAE	0.033239	0.033232	0.033797
R^2	0.472392	0.472385	0.673452
MSE	0.003218	0.003220	0.002934
RMSE	0.047429	0.047428	0.054167

Table 1: R^2 comparison for several ML regressors

Their evaluation has been obtained over the test dataset. All of them are well-known in regression problems. However, the most used in practice is R-squared, given its direct relationship to the generalization power of systems [57].

Consequently, based on Table 1, the prototype chosen for the final implementation is MLP-R. As with the previous two systems, a corresponding object with the typical Pytorch[®] "`__init__`" and "`forward`" methods has also been created (detailed in the next section). ~~At a more abstract level, the five outputs are not independent; instead, they are integrated into a single MLP. This enables the~~

modeling of potential cross-dependencies among output variables, which the two previous systems ignore.

To further assess the robustness and stability of the selected MLP regressor, a 10-fold cross-validation procedure was performed.

FOLD (CV)	MAE	R ²	MSE	RMSE
0	0.041997	0.551366	0.003836	0.061937
1	0.030943	0.722502	0.002227	0.047191
2	0.033151	0.717506	0.002662	0.051591
3	0.033866	0.637492	0.002914	0.053983
4	0.033792	0.679532	0.002765	0.052587
5	0.032826	0.654369	0.003018	0.054937
6	0.031769	0.692778	0.002488	0.049875
7	0.041650	0.610464	0.004424	0.066516
8	0.035136	0.720755	0.003473	0.058931
9	0.030784	0.715950	0.002169	0.046575
Average	0.034591	0.670271	0.002998	0.054413

Table 2: Cross Validation MLP-Regressor Results

Tables 1 and 2 show R² values considerably far from 1. This is partly due to it measures the degree of generalization across multiple outputs, i.e., an average behavior. Therefore, it is expected that the magnitude of this index will be lower compared to what would be obtained with a single output. On the other hand, in the case of the MLP-R, a previously discussed factor comes into play: the outputs are not independent, since each of them is a property of the same material. This is reflected in the MLP architecture, where a given weight can influence multiple outputs, except, obviously, for those in the output layer. This decline in R² has been documented and theoretically justified in prior studies [58]

In summary, the MLP-R was chosen as the final prototype not only for its improved R² performance, but also for its greater consistency with the problem’s physics, grounded in the assumption that the outputs are correlated.

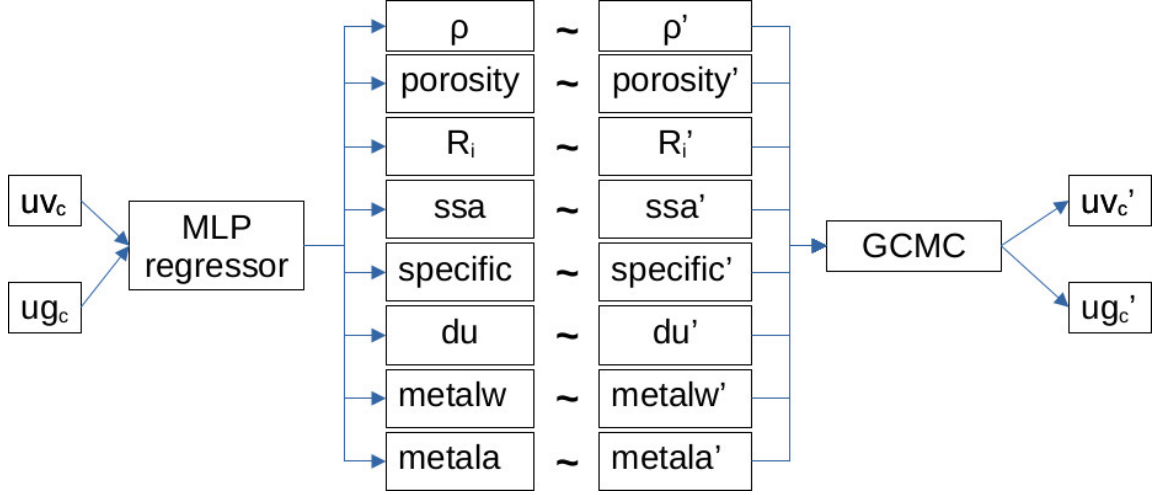


Fig. 2. ANN output as starting point to search a MOFs solution

3.1. Inverse Mapping and Identification of Candidate MOFs

After determining that the MLP Regressor produces the best results, it can be used as an inverse mapping tool. As mentioned before, the main goal of this system should be to calculate the structural parameters of a MOF (outputs) for given ug_c and uv_c (inputs). The prototype's response likely doesn't correspond to a real MOF that will allow us to build a practical storage tank for a FCV. However, a region around the outputs could bound the range of the structural parameters of the MOF where a solution is sought (see Fig. 2) or a MOF whose usable capacities are close to the goals can be found.

In this sense, the ANN should be understood as a guiding tool to delimit promising regions of the structural space, rather than as a direct generator of experimentally realizable materials.

Based on the predicted structural parameters, MOFs with similar characteristics were identified within the CCDC database. Grand Canonical Monte Carlo simulations were then performed for these selected materials at 298.15 K and 25 MPa in order to evaluate their hydrogen storage performance. The corresponding structural parameters and usable storage capacities are reported in Table 5.

To illustrate the application of the inverse mapping strategy, two representative target pairs of usable gravimetric and volumetric capacities were selected. Table 3 summarize the MOFs structural parameters as output according to the two sets of given (or input) hydrogen ug_c and uv_c ,

ug_c	uv_c	density	porosity	R_i	SSA	SPV	du	metala	metalw
2.75	0.04	1.117129	0.411969	6.313358	1364.668457	0.283800	123.282326	46.518978	17.160326
5.50	0.04	0.519389	0.568643	7.746031	3891.882812	0.985582	198.224533	25.361586	4.957312

Table 4: MOF’s prediction provided MLP-R

obtained using and omitting the Hold-Out methodology.

Hold-Out Prototype Outputs						
ug_c	uv_c	density	porosity	R_i	SSA	SPV
2.75	0.04	0.597	0.413	5.69	3562	0.567
5.50	0.04	0.386	0.673	8.37	5166	2.125
No Hold-Out Prototype Outputs						
ug_c	uv_c	density	porosity	R_i	SSA	SPV
2.75	0.04	1.058	0.344	4.83	2772	0.282
5.50	0.04	0.416	0.645	8.03	4810	1.717

Table 3: Hold-Out and No Hold-Out Prototype Outputs

0 2.75 0.04 1.117129 0.411969 6.313358 1364.668457 0.283800 123.282326 46.518978 17.160326
1 5.50 0.04 0.519389 0.568643 7.746031 3891.882812 0.985582 198.224533 25.361586 4.957312

The ANN predicted the structural parameters shown in Table 3. We have inspected the MOFs of the CCDC database and have looked for MOFs whose structural parameters match or are close to the values predicted by the ANN and presented in that Table. We have found some MOFs in the CCDC database whose structural parameters are close, although not the five parameters are close to the predicted ones. We have performed GCMC simulations of those selected MOFs at 298.15 K and 25 MPa. The CCDC identifiers of those MOFs and their structural parameters and usable storage capacities are shown in Table 5.

Regarding the computational volume required to train the proposed ANN systems, the elapsed time for 200 epochs of the first system was less than three minutes on a Linux virtual machine with eight cores of an Intel(R) Xeon(R) Gold 6128 CPU @ 3.40 GHz. It shows that a GPU is not crucial for the present work.

Table 5: Structural parameters and usable storage capacities at 298.15 and 25 MPa of the selected MOFs. The reported parameters include framework density, (ρ , kg/L), porosity, (Prs., dimensionless), largest pore radius, (R_i , Å), specific surface area (SSA, m²/g), specific pore volume (SPV, cm³/g), usable gravimetric capacity (ug_c , wt.%), and usable volumetric capacity (uv_c , kg/L).

MOF	Properties	Prs.	ρ	R_i	SSA	SPV	ug_c	ug_c
	alejae	0.694	0.283	11.20	4795.04	2.54398	0.0162	5.40
	bazfuf01	0.566	0.357	11.25	5191.53	1.67818	0.0157	4.20
	bazfuf	0.567	0.355	11.33	5140.92	1.74437	0.0158	4.25
	bibxob	0.541	0.408	11.07	4641.18	1.51819	0.0159	3.75
	bihbax	0.688	0.334	9.96	4395.86	2.03445	0.0173	4.92
	bognes	0.429	0.415	10.31	5648.74	1.01184	0.0136	3.18
	dajwet	0.621	0.295	14.88	4985.43	2.50898	0.0154	4.98
	dimbot	0.746	0.430	14.99	1903.14	1.72163	0.0166	3.72
	ecokaj	0.682	0.333	8.97	3838.23	2.03829	0.0153	4.39
	ehijah	0.574	0.385	10.38	4589.81	1.66174	0.0152	3.80
	fe-pbpta	0.669	0.411	10.09	4336.69	1.65417	0.0179	4.17
	ipupoa	0.551	0.403	10.58	4794.61	1.33786	0.0154	3.67
	JLU-MONT1_JEYTEP	0.802	0.192	14.53	6034.91	4.47919	0.0162	7.77
	JLU-SOF2_KODPIF	0.749	0.232	14.12	4512.98	3.28028	0.0153	6.20
	JLU-SOF3_KODPIF01	0.747	0.151	11.54	7243.99	5.56697	0.0164	9.83
	lejcio	0.612	0.326	10.35	5125.64	1.96099	0.0156	4.56
	omusas	0.601	0.361	7.83	3822.92	1.66278	0.0140	3.75
	opawot	0.566	0.368	7.03	4834.32	1.35793	0.0153	4.00
	uqoluj01	0.613	0.357	10.14	4834.06	1.62163	0.0156	4.19
	vetmis	0.609	0.321	5.11	5462.80	2.30117	0.0155	4.59
	xahput	0.755	0.186	11.83	5870.20	4.08809	0.0162	8.01
	xahqaa	0.779	0.176	12.31	5832.99	4.32	0.0164	8.52
	xanlij	0.692	0.231	13.73	5706.56	3.03037	0.0154	6.25

3.2. Illustrative Comparison with a Commercial FCV

To illustrate the potential system-level implications of the obtained storage capacities, a simplified comparison with a commercial hydrogen fuel cell vehicle was carried out. The 2024 Toyota Mirai was selected as a reference system due to the availability of public technical data.

The hydrogen storage system of the 2024 Toyota Mirai, a hydrogen fuel cell vehicle, consists of tanks with a total volume of 142 litres, stores 5.6 kg of hydrogen at 70 MPa and the vehicle has a driving range of 402 miles (647 km). If a selected MOF is incorporated into an Adsorbed Gas, AG, storage system within the same vehicle and with a total volume of $2.5 \times 142 = 355$ litres, the driving range in km of the AG vehicle will be $ug_c \times 355 \text{ litres} \times 647 \text{ km} / 5.6 \text{ kg}$, where ug_c is in kg/L units. That driving range is shown in Table 6.

Table 6: Usable volumetric capacity (ug_c , kg/L) at 298.15 K and 25 MPa, and estimated driving range (in km) of the selected MOFs. The calculated ranges are compared with the performance of the 2024 Toyota Mirai FCV.

MOF	ug_c	Driving Range
alejae	0.0162	664
bazfuf01	0.0157	644
bazfuf	0.0158	648
bibxob	0.0159	652
bihbax	0.0173	710
dajwet	0.0154	632
dimbot	0.0166	681
ecokaj	0.0153	628
ehijah	0.0152	623
fe-pbpta	0.0179	734
ipupoa	0.0154	632
JLU-MONT1_JEYTEP	0.0162	664
JLU-SOF2_KODPIF	0.0153	628
JLU-SOF3_KODPIF01	0.0164	673
lejcio	0.0156	640
omusas	0.0140	574
opawot	0.0153	628
uqoluj01	0.0156	640
vetmis	0.0155	636
xahput	0.0162	664
xahqaa	0.0164	673
xanlij	0.0154	632
Toyota Mirai (2024)	0.0390	647

Hence, most of the selected MOFs have a usable gravimetric capacity that reaches the DOE 2025 target and a driving range, with a 355 litres storage tank system, that reaches or is close to the driving range of an existing FCV, the 2024 Toyota Mirai.

Hence, most of the selected MOFs exhibit usable gravimetric capacities that reach the DOE 2025 target. However, their usable volumetric capacities remain below the DOE requirement. For this reason, the comparison with the 2024 Toyota Mirai is presented only as an illustrative system-level analysis. Using an adsorbed-gas storage system operating at 25 MPa, an internal tank volume of 355 litres (i.e., approximately 2.5 times the volume of the compressed-gas system at 70 MPa) would be required to achieve a driving range comparable to that of the Mirai. This result highlights that, within the explored structural space, the model does not identify MOFs that simultaneously satisfy both the gravimetric and volumetric DOE targets at moderate pressure.

Nevertheless, operating at lower pressure (25 MPa instead of 70 MPa) offers practical advantages, including reduced system cost, improved safety margins, enhanced durability, and lower maintenance requirements.

Future extensions of this approach may explore broader structural spaces or alternative operating conditions to further improve volumetric performance.

4. Conclusions

Leveraging Artificial Intelligence, an ANN-based system was designed to act as a nonlinear function estimator for hydrogen storage problems characterized by complex structure-property relationships. From a computational standpoint, it can be categorized as an intelligent system within the field of Soft Computing, given its relatively low resource consumption. The Multilayer Perceptron (MLP) has yielded robust and reliable results as a nonlinear regressor [57].

Using this model, a specific region in the MOF space has been predicted where materials are likely to meet the DOE hydrogen storage targets. The prediction is based on key structural descriptors such as porosity (Prs), density (ρ), average pore radius (Ri), specific surface area (SSA), and specific pore volume (SPV). These parameters strongly influence both the gravimetric and volumetric hydrogen storage capacities. This approach can help experimental researchers focus their synthesis efforts on MOFs showing combinations of these structural properties that are more likely to achieve the DOE requirements. Furthermore, Grand Canonical Monte Carlo (GCMC) simulations performed on MOFs close to this predicted region ~~confirmed that several of them satisfy three out of the five DOE parameters.~~ confirmed that several of them meet three DOE-related performance metrics explicitly addressed in this work, namely: (i) the usable gravimetric capacity, (ii)

the usable volumetric capacity, and (iii) the associated driving range derived from these capacities under the assumed tank conditions.

Remarkably, even considering only three of these structural properties, the usable gravimetric capacity equals or exceeds the DOE target. Although the volumetric capacity reaches about half the required value, increasing the tank volume by roughly 2.5 times would provide a driving range comparable to that of the Toyota Mirai 2024.

Notably, even when considering only three structural descriptors, the usable gravimetric capacity of several MOFs equals or exceeds the DOE 2025 target. In contrast, the usable volumetric capacity remains approximately half of the required value. As a consequence, achieving a driving range comparable to that of the 2024 Toyota Mirai at 25 MPa would require a storage volume about 2.5 times larger than that of the 70 MPa compressed-gas system. This finding clarifies that the proposed approach does not meet the DOE volumetric target at moderate pressure, but it delineates realistic trade-offs between storage volume, operating pressure, and system-level performance.

Lower-pressure operation provides relevant system-level advantages, including reduced cost, improved durability, enhanced safety, and lower maintenance requirements, which are important considerations for the practical deployment of hydrogen storage technologies.

The usable gravimetric (ug_c) and volumetric (uv_c) capacities are calculated based on the mass of adsorbed hydrogen and the framework only, following DOE conventions. The total system mass, including tanks or supports, is not included in these calculations.

5. Acknowledgments

This work was supported by financial aid from the Spanish Ministry of Science and Innovation (Project PGC2018-093745-B-I00), the Regional Government of Castilla y León (Project VA124G18), and the University of Valladolid. We are sincerely thankful for their contribution, which made this study possible. We also wish to recognize the use of the computing resources offered by the Data Processing Center at the Science Park of the University of Valladolid.

References

- [1] L. Zhang, C. Jia, F. Bai, W. Wang, S. An, K. Zhao, Z. Li, J. Li, H. Sun, A comprehensive review of the promising clean energy carrier: Hydrogen production, transportation, storage, and utilization (HPTSU) technologies, *Fuel* 355 (2024) 129455. doi:10.1016/j.fuel.2023.129455.

- [2] S. Chakraborty, S. K. Dash, R. M. Elavarasan, A. Kaur, D. Elangovan, S. T. Meraj, P. Kasiathan, Z. Said, Hydrogen energy as future of sustainable mobility, *Front. Energy Res.* 10 (2022) 893475. doi:10.3389/fenrg.2022.893475.
- [3] P. Jena, Materials for hydrogen storage: Past, present, and future, *J. Phys. Chem. Lett.* 2 (3) (2011) 206–211. doi:10.1021/jz1015372.
- [4] European Environment Agency, Transport and mobility, <https://www.eea.europa.eu/en/topics/in-depth/transport-and-mobility>, Accessed January 2, 2026 (2024).
- [5] Office of Energy Efficiency & Renewable Energy, Fuel Cell Technologies Office, DOE technical targets for onboard hydrogen storage for light-duty vehicles, <https://www.energy.gov/eere/fuelcells/doe-technical-targets-onboard-hydrogen-storage-light-duty-vehicles>, Accessed January 2, 2026 (2018).
- [6] G. Assoualaye, A. Tom, N. Djongyang, Monte Carlo study of hydrogen adsorption by MOF-5 doped with cobalt at ambient temperature and pressure, *SN Appl. Sci.* 2 (2020) 1815. doi:10.1007/s42452-020-03627-9.
- [7] M. D. Allendorf, Z. Hulvey, T. Gennett, A. Ahmed, T. Autrey, J. Camp, E. S. Cho, H. Furukawa, M. Haranczyk, M. Head-Gordon, S. Jeong, A. Karkamkar, D.-J. Liu, J. R. Long, K. R. Meihaus, I. H. Nayyar, R. Nazarov, D. J. Siegel, V. Stavila, J. J. Urban, S. P. Veccham, B. C. Wood, An assessment of strategies for the development of solid-state adsorbents for vehicular hydrogen storage, *Energy Environ. Sci.* 11 (2018) 2784–2812. doi:10.1039/C8EE01085D.
- [8] C. Chu, K. Wu, B. Luo, Q. Cao, H. Zhang, Hydrogen storage by liquid organic hydrogen carriers: Catalyst, renewable carrier, and technology – a review, *Carbon Resour. Convers.* 6 (4) (2023) 334–351. doi:10.1016/j.crcon.2023.03.007.
- [9] D. P. Broom, C. J. Webb, K. E. Webb, P. A. Parilla, T. Gennett, C. M. Brown, R. Zacharia, E. Tylianakis, E. Klontzas, G. E. Froudakis, T. A. Steriotis, P. N. Trikalitis, D. L. Anton, B. Hardy, D. Tamburello, C. Corgnale, B. A. van Hassel, D. Cossement, R. Chahine, M. Hirscher, Outlook and challenges for hydrogen storage in nanoporous materials, *Appl. Physics A* 122 (2021) 151. doi:10.1007/s00339-016-9651-4.

- [10] P. Breeze, Hydrogen energy storage, Elsevier, London, 2018, Ch. 8, pp. 69–77.
- [11] H. T. Hwang, A. Varma, Hydrogen storage for fuel cell vehicles, *Curr. Opin. Chem. Eng.* 5 (2014) 42–48. doi:10.1016/j.coche.2014.04.004.
- [12] F. Ding, B. I. Yakobson, Challenges in hydrogen adsorptions: from physisorption to chemisorption, *Front. Phys.* 6 (2011) 142–150. doi:10.1007/s11467-011-0171-6.
- [13] A. Granja-DelRío, I. Cabria, Grand canonical Monte Carlo simulations of hydrogen and methane storage capacities of two novel Al-nia MOFs at room temperature, *Int. J. Hydrogen Energy* 50 (2024) 685–696. doi:10.1016/j.ijhydene.2023.08.023.
- [14] M. López, M. B. Torres, I. Cabria, Grand Canonical Monte Carlo simulations of the hydrogen storage capacities of zeolite-templated carbon schwarzites at room temperature, *Int. J. Hydrogen Energy* 71 (2024) 1363–1372. doi:10.1016/j.ijhydene.2024.05.256.
- [15] I. Cabria, A. Lebon, M. B. Torres, L. J. Gallego, A. Vega, Hydrogen storage capacity of Li-decorated borophene and pristine graphene slit pores: a combined ab-initio and quantum-thermodynamic study, *Appl. Surf. Sci.* 562 (2021) 150019. doi:10.1016/j.apsusc.2021.150019.
- [16] L. T. Glasby, P. Z. Moghadam, Hydrogen storage in mofs: machine learning for finding a needle in a haystack, *Patterns* 2 (7) (2021) 100305. doi:10.1016/j.patter.2021.100305.
- [17] K. Salehi, M. Rahmani, S. Atashrouz, Machine learning assisted predictions for hydrogen storage in metal–organic frameworks, *Int. J. Hydrogen Energy* 48 (85) (2023) 33260–33275. doi:10.1016/j.ijhydene.2023.04.338.
- [18] H. Qiu, Y. Xia, C. Xiang, F. Xu, L. Sun, Y. Zou, Prediction of hydrogen storage in metal–organic frameworks using catboost-based approach, *Int. J. Hydrogen Energy* 79 (2024) 952–961. doi:10.1016/j.ijhydene.2024.07.078.
- [19] Z.-H. Cao, X.-Q. Bian, J. Chen, J.-Y. Zhang, W. Fu, Predicting hydrogen storage in metal–organic frameworks using a novel hybrid machine learning model, *Int. J. Hydrogen Energy* 145 (2025) 401–411. doi:10.1016/j.ijhydene.2025.06.112.
- [20] A. Ahmed, D. J. Siegel, Predicting hydrogen storage in MOFs via machine learning, *Patterns* 2 (7) (2021) 100291. doi:10.1016/j.patter.2021.100291.

- [21] H. Wang, M. Wang, Y. Yin, Z. Qu, A universal structure of neural network for predicting heat, flow and mass transport in various three-dimensional porous media, *International Journal of Heat and Mass Transfer* 241 (2025) 126688. doi:10.1016/j.ijheatmasstransfer.2025.126688.
- [22] S.-Y. Lee, J.-H. Lee, Y.-H. Kim, J.-W. Kim, K.-J. Lee, S.-J. Park, Recent progress using solid-state materials for hydrogen storage: A short review, *Processes* 10 (2022) 304. doi:10.3390/pr10020304.
- [23] S. P. Shet, S. S. Priya, K. Sudhakar, M. Tahir, A review on current trends in potential use of metal-organic framework for hydrogen storage, *Int. J. Hydrogen Energy* 46 (2021) 11782–11803. doi:10.1016/j.ijhydene.2021.01.020.
- [24] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 2nd Edition, O'Reilly Media, 2019.
- [25] I. Cabria, A fortran code to perform Grand Canonical Monte Carlo simulations of the gas storage capacities of solid materials, <https://github.com/ivancabria/mcmgs/tree/master>, <https://zenodo.org/records/15708499>, Last Version: 1.2, December 5, 2025 (10 2025). doi:10.5281/zenodo.15708498.
- [26] Cambridge Crystallographic Database Centre, CCDC, Structural chemistry data, software, and insights, www.ccdc.cam.ac.uk, Accessed January 2, 2026 (2025).
- [27] N. Metropolis, The Beginning of the Monte Carlo Method, *Los Alamos Science* 15 (1987) 125–130.
- [28] D. Caviedes, I. Cabria, Grand Canonical Monte Carlo simulations of the hydrogen storage capacities of slit-shaped pores, nanotubes and torusenes, *Int. J. Hydrogen Energy* 47 (2022) 11916 – 11928. doi:10.1016/j.ijhydene.2022.01.229.
- [29] A. Granja-DelRío, I. Cabria, Exploring the hydrogen and methane storage capacities of novel DUT MOFs at room temperature: A grand canonical Monte Carlo simulation study, *Int. J. Hydrogen Energy* 54 (2024) 665–677. doi:10.1016/j.ijhydene.2023.11.258.
- [30] G. Soave, Equilibrium constants from a modified Redlich-Kwong equation of state, *Chem. Eng. Sci.* 27 (1972) 1197–1203. doi:10.1016/0009-2509(72)80096-4.

- [31] L. Zhou, Y. Zhou, Determination of compressibility factor and fugacity coefficient of hydrogen in studies of adsorptive storage, *Int. J. Hydrogen Energy* 26 (2001) 597–601.
- [32] X.-H. Xu, Y.-Y. Duan, Z. Yang, Crossover volume translation Soave-Redlich-Kwong equation of state for fluids, *Ind. Eng. Chem. Res.* 51 (2012) 6580–6585.
- [33] J. E. Lennard-Jones, On the determination of molecular fields, *Proc. Roy. Soc. (London) A* 106 (1924) 463–477. doi:10.1098/rspa.1924.0082.
- [34] R. J. Good, C. J. Hope, New combining rule for intermolecular distances in intermolecular potential functions, *J. Chem. Phys.* 53 (1970) 540–543. doi:10.1063/1.1674022.
- [35] D. Berthelot, Sur le mélange des gaz, *Comptes rendus hebdomadaires des séances de l’Académie des Sciences* 126 (1898) 1703.
- [36] A. Granja-DelRío, I. Cabria, Insights into hydrogen and methane storage capacities: Grand Canonical Monte Carlo simulations of SIGSUA, *J. Chem. Phys.* 204 (2024) 154712. doi:10.1063/5.0193291.
- [37] A. Granja-DelRío, I. Cabria, Grand canonical monte carlo simulations of the hydrogen and methane storage capacities of jlu-mof120 and jlu-mof121, *Int. J. Hydrogen Energy* 61 (2024) 57–72.
- [38] A. Granja-DelRío, I. Cabria, Grand Canonical Monte Carlo simulations of the Hydrogen and Methane storage capacities of a novel Co-MOF, *Results in Surfaces and Interfaces* 18 (2025) 100442. doi:10.1016/j.rsurfi.2025.100442.
- [39] D. Dubbeldam, S. Calero, D. E. Ellis, R. Q. Snurr, RASPA: molecular simulation software for adsorption and diffusion in flexible nanoporous materials, *Molecular Simulation* 42 (2) (2016) 81–101. doi:10.1080/08927022.2015.1010082.
- [40] M. Haranczyk, Zeo++: an open source software for performing high-throughput geometry-based analysis of porous materials and their voids, <http://www.zeoplusplus.org>, Last version: 0.3, June 20, 2017. Accessed January 2, 2026 (2017).
- [41] P. Ramirez-Vidal, G. Sdanghi, A. Celzard, V. Fierro, High hydrogen release by cryo-adsorption and compression on porous materials, *International Journal of Hydrogen Energy* 47 (14) (2022) 8892–8915. doi:10.1016/j.ijhydene.2021.12.235.

- [42] M. Schlichtenmayer, M. Hirscher, The usable capacity of porous materials for hydrogen storage, *Appl. Physics A* 122 (2016) 379. doi:10.1007/s00339-016-9864-6.
- [43] J. F. T. Hastie, R. Tibshirani, *The Elements of Statistical Learning*, 2nd Edition, Springer, New York, 2009.
- [44] C. M. Bishop, *Pattern Recognition and Machine Learning*, Information Science and Statistics, Springer, New York, 2006.
- [45] T. M. Mitchell, *Machine Learning*, McGraw-Hill, New York, 1997.
- [46] M. A. H. I. H. Witten, E. Frank, *Data Mining: Practical Machine Learning Tools and Techniques*, 3rd Edition, Morgan Kaufmann Publishers, Burlington, MA, 2011.
- [47] G. James, D. Witten, T. Hastie, R. Tibshirani, *An Introduction to Statistical Learning*, 2nd Edition, Springer, 2021, states: "The most common measure of fit for regression models is R^2 ... it is ubiquitously used in practice and academic literature."
- [48] Y.-W. Chang, C.-J. Hsieh, K.-W. Chang, M. Ringgaard, C.-J. Lin, Training and testing low-degree polynomial data mappings via linear svm, *Journal of Machine Learning Research* 11 (2010) 1471–1490.
- [49] F. Nie, W. Zhu, X. Li, Decision tree svm: An extension of linear svm for non-linear classification, *Neurocomputing* 401 (2020) 153–159. doi:<https://doi.org/10.1016/j.neucom.2019.10.051>. URL <https://www.sciencedirect.com/science/article/pii/S0925231219314304>
- [50] S. Raschka, J. Patterson, C. Nolet, Machine learning in python: Main developments and technology trends in data science, machine learning, and artificial intelligence, *Information* 11 (4) (2020) 193. doi:10.3390/info11040193. URL <https://www.mdpi.com/2078-2489/11/4/193>
- [51] C. M. Bishop, *The multi-layer perceptron*, Oxford University Press, Oxford, UK, 1995, Ch. 4, pp. 116–163.
- [52] S. Haykin, *Neural Networks: A Comprehensive Foundation*, 2nd Edition, Prentice Hall, Upper Saddle River, NJ, 1999.

- [53] V. N. Vapnik, The Nature of Statistical Learning Theory, 2nd Edition, Springer-Verlag, 1999.
- [54] L. Bottou, F. E. Curtis, J. Nocedal, Optimization methods for large-scale machine learning, SIAM Review 60 (2018) 223–311.
- [55] D. Choi, M. Song, A survey on optimization algorithms for deep learning methods, Journal of Information Processing Systems 16 (2020) 1267–1284.
- [56] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, <https://arxiv.org/abs/1412.6980>, poster at 3rd Int. Conf. Learning Representations (ICLR), San Diego, CA, USA, May 2015 (2017). arXiv:1412.6980.
- [57] I. Goodfellow, Y. Bengio, A. Courville, Deep Learning, Adaptive Computation and Machine Learning series, The MIT Press, Cambridge, Massachusetts, USA, 2016.
- [58] R. A. Maronna, R. D. Martin, V. J. Yohai, M. Salibián-Barrera, Linear regression 2, in: Robust Statistics: Theory and Methods (with R), 2nd Edition, Wiley, Hoboken, NJ, 2019, Ch. 5, pp. 115–191.